DO NOW — RECALL

Lesson 097 · Little Man Computer: Branches and Loops

Name: _____ Class: _____ Date: _____

1. What does BRA do in LMC?

2. When does BRZ branch?

3. What does LDA COUNT do?

4. Stack Push2,Push5,Pop,Push8: final bottom-to-top?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Branches unconditionally to a label.

2. Model answer

When accumulator equals zero.

3. Model answer

Loads COUNT's mailbox contents into ACC.

4. Model answer

[2,8].

By the end, you can…

• BRA always changes which instruction is fetched next; BRZ branches only when ACC is zero.

• The loop's memory variable COUNT changes after SUB ONE and STA COUNT. Reload it with LDA COUNT at each iteration.

• Trace ACC, COUNT, branch decision and outputs. Stopping before OUT when COUNT=0 prevents printing zero.

Vocabulary — know these terms

BRA — Always branch to label.

BRZ — Branch if ACC zero.

label — Symbolic instruction/memory address.

loop — Repeated control flow.

termination — Stop condition for loop.

NEW LEARNING

Read, annotate and explain

Observe

BRA always changes which instruction is fetched next; BRZ branches only when ACC is zero.

Apply

The loop's memory variable COUNT changes after SUB ONE and STA COUNT. Reload it with LDA COUNT at each iteration.

Evaluate

Trace ACC, COUNT, branch decision and outputs. Stopping before OUT when COUNT=0 prevents printing zero.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Start COUNT=2 instead of3. What gets printed and why?

Follow the reasoning

Outputs2 then1; COUNT reaches0, and BRZ branches before OUT.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace COUNT DAT3 loop outputs and COUNT after each SUB. [6]

Question 2

Change COUNT DAT3 to COUNT DAT2. State outputs and explain BRZ. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A pupil replaces BRZ END with BRA END at LOOP. Explain effect and repair. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

BRZ END always jumps to END regardless of ACC.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.